

Lab-I
National Institute of Technology Silchar

Subject Code: CS-5104
Semester: 1st M.Tech.
Section: All

Subject: Computer System Lab
Department: CSE
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Teaching Assistant (TA) will ensure that you have completed your tasks or not. The TA will record the performance for future grading. You need to write the programs and/or outputs on your lab copy as directed in the questions.

1. Write a C program to create a child process using fork(). Print the PID and PPID of both the parent and child processes.
2. How many times will Hello be printed by the following program?

```
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>

int main()
{
    for (int i=0; i<2; i++)
        fork();
    printf("Helloo\n");
}
```

3. Consider the C program given below. Determine:

- i) the total number of processes, including the original parent process, and
- ii) the total number of child processes created.

Also, illustrate the process creation by drawing an appropriate process tree.

```
(a) #include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
int main()
{
    for (int i=0; i<4; i++)
        fork();
}
```

```
(b) #include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
int main()
{
    for (int i=0; i<3; i++)
    {
        fork();
    }
}
```

```

        fork ();
    }
}

(c) #include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
int main()
{
    for (int i = 0; i < 2; i++)
    {
        for (int j = 0; j < 2; j++)
        {
            fork ();
        }
    }
}

```

4. Explain the following phenomenon of the program-

```

#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>
int main()
{
    pid_t pid = fork ();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }
    else if (pid == 0)
    {
        printf("Before parent terminates:\n");
        printf("Child PID=%d, Parent PID=%d\n", getpid (),
            getppid());
        sleep(5);
        printf("After parent terminates:\n");
        printf("Child PID=%d, Parent PID=%d\n", getpid (),
            getppid());
    }
    else
    {
        printf("Parent PID = %d\n", getpid());
        exit(0);
    }
    return 0;
}

```

5. Explain the child process during the 100 seconds

```

#include <stdio.h>

```

```

#include <unistd.h>
#include <stdlib.h>
int main()
{
    pid_t pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }
    else if (pid == 0)
    {
        printf("Child process is terminating.\n");
        printf("Child PID = %d\n", getpid());
        exit(0);
    }
    else
    {
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
        sleep(100);
        printf("Parent is terminating now.\n");
    }

    return 0;
}

```

6. Explain the following program-

```

(a) #include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
int main()
{
    int n, i;
    int sum = 0;
    pid_t pid;

    printf("Enter the value of n: ");
    scanf("%d", &n);

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }
    else if (pid == 0)
    {
        for (i = 1; i <= n; i++)

```

```

        {
            if (i % 2 != 0)
                sum += i;
        }
        printf("Child: Sum of odd numbers = %d\n", sum);
    }
    else
    {
        for (i = 1; i <= n; i++)
        {
            if (i % 2 == 0)
                sum += i;
        }
        printf("Parent: Sum of even numbers = %d\n", sum);
    }
    return 0;
}

```

```

(b) #include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
int main()
{
    int x = 10;
    pid_t pid;

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }
    else if (pid == 0)
    {
        x = x + 5;
        printf("Child process:\n");
        printf("PID = %d, x = %d\n", getpid(), x);
    }
    else
    {
        x = x + 10;
        printf("Parent process:\n");
        printf("PID = %d, x = %d\n", getpid(), x);
    }

    return 0;
}

```

***** END *****